

SSCR-POWERED P2P LENDING PLATFORM

FULL TECHNICAL SYSTEM REPORT (DETAILED VERSION)

1. EXECUTIVE SUMMARY

This report proposes a complete technical architecture for a Self-Sovereign Credit Rating (SSCR) based peer-to-peer lending platform integrating decentralized identity, verifiable credentials, machine learning credit scoring, asset verification,

loan origination, marketplace publishing, investor funding, and repayment automation.

The platform ensures:

- Trustless identity verification
- Fraud-proof borrower credentials
- Transparent credit scoring
- Reduced operational cost
- Lower borrower interest rates
- Higher lender returns than bank deposits
- Secure & auditable loan lifecycle

2. SYSTEM OBJECTIVES

- Create a decentralized credit evaluation system using Verifiable Credentials (VCs)
- Enable borrowers to request loans with verified identity & assets
- Allow investors to fund projects based on transparent risk scores
- Automate risk scoring using ML + SSCR logic
- Maintain secure on-chain proofs of verification
- Build a compliant financial system for Nepal's regulatory structure

3. SYSTEM STAKEHOLDERS

3.1 Borrower

- Receives loan
- Provides verifiable identity & financial credentials
- Maintains an updated SSCR score

3.2 Originator

- Performs checks
- Verifies credentials & assets
- Publishes loan requests to marketplace

3.3 Capital Provider (Investor)

- Funds loan projects
- Earns interest based on risk band

3.4 Verifiers / Issuers

- Department of Survey (Land)
- PAN/TAX Authorities
- Banks & MFIs (Financial history)
- Telecom (Address verification)
- Employer (Income verification)

3.5 Platform Operator

- Manages technical infrastructure
- Ensures compliance with NRB regulations

4. SYSTEM ARCHITECTURE OVERVIEW

The platform uses a hybrid architecture:

1. ****Decentralized Layer****

- Hyperledger Indy / Aries for DID & VCs

- Credential schemas stored on-chain
- Revocation registry managed by issuers

2. **Off-chain Backend Services**

- Loan origination service
- Risk scoring engine
- Asset verification API
- Marketplace service
- Repayment & EMI service

3. **Frontend Applications**

- Borrower mobile app
- Originator web dashboard
- Investor dashboard
- Verifier portal (government, banks)

4. **Database Systems**

- PostgreSQL for main DB
- Redis for cache + rate limiting
- Elasticsearch for marketplace search

5. IDENTITY & CREDENTIAL LAYER

5.1 DID Architecture

Each user receives:

- A decentralized identifier
- Stored in mobile wallet
- Verified using Hyperledger Indy

Structure:

did:indy:region:unique_id

5.2 Verifiable Credentials (VCs)

Borrower receives cryptographically signed credentials:

- National Identity VC
- KYC VC
- Address VC
- Employment VC
- Income VC
- Land Ownership VC
- Bank Statement VC
- PAN & Tax VC
- Loan Repayment History VC (generated internally)

Each VC includes:

- Issuer DID
- Schema
- Expiration
- Revocation status

5.3 Zero-Knowledge Proofs (ZKP)

Borrower can prove:

- Income is above a certain threshold
- without revealing actual salary.

6. ASSET VERIFICATION & LAND REGISTRY

The platform integrates with Nepal's **Department of Survey**:

1. Borrower submits land certificate
2. System extracts:

- Parcel ID

- Owner name

- Area

- Valuation

3. System verifies:

- Ownership match

- Encumbrance (existing loans)

- Disputes

4. Creates a ****Land Ownership VC****

Collateral-to-loan ratio is calculated:

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Collateral Score = Land Value / Loan Requested

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7. LOAN ORIGINATION WORKFLOW

1. Borrower applies for a loan

2. Submits:

- Identity VC

- Income VC

- Land VC (optional collateral)

- Employment VC

- Bank Statement VC

3. Originator Backend validates credentials

4. Risk Engine computes SSCR score

5. Originator sets:

- Loan Amount
- Tenure
- Base Interest
- Risk Premium

6. Project published on marketplace

8. SSCR RISK ENGINE

The engine uses:

- Financial Stability Score (income, savings, debt)
- Collateral Score (land, assets)
- Behavior Score (late payments)
- Identity Trust Score (VCs verified)
- ML Score (default prediction)

Final SSCR Score:

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$$\text{SSCR} = 0.25 \times \text{Financial} + 0.20 \times \text{Collateral} + 0.15 \times \text{Behavior} \\ + 0.10 \times \text{Identity} + 0.30 \times \text{ML\_Score}$$

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Range:

- 800–900 → A+ (Very Low Risk)
- 700–799 → A (Low Risk)
- 600–699 → B (Moderate Risk)
- 500–599 → C (High Risk)
- 350–499 → D (Very High Risk)

9. INTEREST RATE LOGIC

Interest = Base Rate + Risk Premium

Where:

- Base Rate ~ 10–12% (market baseline)
- Risk Premium depends on SSCR

Example:

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A+ : 10% + 1%

A : 10% + 3%

B : 10% + 5%

C : 10% + 8%

D : 10% + 12%

...

Ensures:

- Borrowers get fair rates
- Investors earn more than bank deposits
- Platform maintains margin through origination fees

10. MARKETPLACE WORKFLOW

1. Originator publishes project

2. Investors filter by:

- SSCR band
- Interest rate
- Tenure
- Collateral

3. Investors commit funds

4. Escrow account holds funds

5. Loan disbursed upon full funding

11. REPAYMENT ENGINE

EMI Calculation:

Standard reducing balance EMI.

Investor Distribution:

Each EMI is split automatically based on investment percentage.

Late Payments:

- Reported back to SSCR ledger
- Immediate score impact
- Risk alerts sent to investors

Defaults:

- Collateral liquidation process triggered
- Default Credential (VC) issued
- Affects future borrowing capacity

12. BLOCKCHAIN INTEGRATION

Blockchain stores only:

- Credential hashes
- Revocation lists
- Risk score proofs
- Transaction attestations

No personal data stored on-chain.

13. TECH STACK

13.1 Backend

- Python (FastAPI / Django)
- PostgreSQL
- Redis
- ElasticSearch
- Kafka for event streaming
- Celery for async tasks

13.2 Frontend

- React (Investor/Admin)
- React Native (Borrower Mobile App)

13.3 Blockchain / SSI

- Hyperledger Indy (DID Registry)
- Hyperledger Aries (VC communication)
- Hyperledger Ursa (crypto primitives)
- Optional: Hyperledger Fabric for loan anchors

13.4 ML Stack

- Python
- XGBoost & LightGBM
- TensorFlow Lite (mobile inference)
- Feature store (Feast)

14. SECURITY MODEL

- All VCs issued are signed with issuer keys
- DID-based authentication replaces passwords

- AES-256 encrypted vault for sensitive data
- Multi-party approval for loan disbursement
- Rate limiting & anomaly detection on borrower activity

15. REGULATORY COMPLIANCE (NEPAL)

- KYC compliance with NRB requirements
- Loan disbursement through regulated BFIs
- Escrow account managed by licensed entity
- Audit logs maintained for 10 years
- GDPR-like privacy compliance

16. FUTURE EXTENSIONS

- Tokenized investment pools
- Real-time credit score updates
- Cross-border investment opportunities
- DeFi lending integration (Polygon / Ethereum)

17. CONCLUSION

This SSCR-powered P2P lending ecosystem introduces a secure, scalable, and transparent financial infrastructure.

It leverages decentralized identity, verifiable credentials, ML-based scoring, and collateral verification to create a next-generation lending system for Nepal.